

§3.3 C^∞ determinant line bundle

We keep the setting above. $\sqrt{2}D \stackrel{\omega^Z \text{ is Kahler}}{=} \sum_{j=1}^{2l} c(e_j) \nabla_{e_j}^F$.

$$2D^2 = \Delta^F + \underbrace{R^\xi(w_j, \bar{w}_j) + \frac{1}{2}(R^\xi + \frac{1}{2}\text{Tr}[R^{T^{1,0}Z}])(w_i, \bar{w}_j)\bar{w}^j \wedge i_{\bar{w}_i}}_{\text{noted by } Q, \text{ zero order part}}$$

where $\Delta^F = -\sum_j (\nabla_{e_j}^F)^2 - \nabla_{\nabla_{e_j}^{TZ} e_j}^F$ Laplacian associated to ∇^F . $2D^2$ is a generalized Laplacian on F , it is a second order elliptic differential operator.

CONFUSION

I TRIED TO APPLY THE EQUATION 1.3.26 OF PROFESSOR'S BOOK, BUT I CANNOT OBTAIN THE HALF AND THE QUATER COEFFICIENTS.

Theorem. *For the compactness of Z , the spectrum $\text{Spec}(D^2)$ of D^2 is discrete and there exists an orthonormal basis $\{\varphi_j\}_{j=1}^\infty$ of $L^2(Z, F)$ such that*

$$D^2 \varphi_j = \lambda_j \varphi_j, \quad 0 \leq \lambda_1 \leq \lambda_2 \leq \dots \rightarrow +\infty.$$

And for each $1 \leq p \leq l$, each $j \in \mathbb{N}^$, the eigenspace is finite dimensionnal, i.e.*

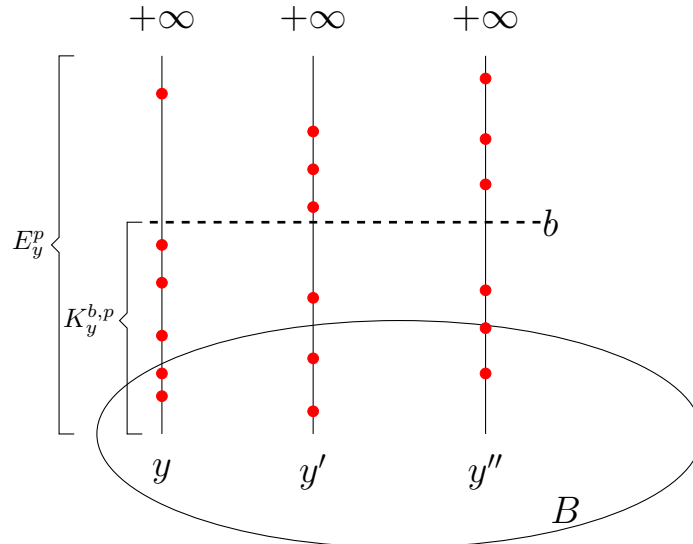
$$\dim \text{Ker}(D^2|_{\Omega^{0,p}} - \lambda_j) < +\infty$$

and the eigenvector is smooth, i.e. $\varphi_j \in C^\infty(Z, F)$.

Back to our problem. For each $y \in B$, let $E_{\mu,y}^p$ be the eigenspace of $D_y^2|_{\Omega^{0,p}(Z_y, \xi|_{Z_y})}$ associated to the eigenvalue μ . For $b > 0$, let $K_y^{b,p} = \bigoplus_{\mu < b} E_{\mu,y}^p$. By the theorem above, $K_y^{b,p}$ is a finite dimensional subspace of $E_y^p = \Omega^{0,p}(Z_y, \xi|_{Z_y})$. Set

$$U^b = \{y \in B \mid b \notin \text{Spec}(D_y^2)\}.$$

See the illustration below, figuring $B \times [0, +\infty)$ and the eigenvalues (red dots).



Theorem. U^b is open, $K^{b,p} = \bigcup_{y \in U^b} K_y^{b,p}$ is a smooth finite dimensional vector bundle on U^b .

Proof. For $y \in U^b$, let $P_y^{b,p} : E_y^p \rightarrow K_y^{b,p}$ be the orthogonal projection. For each $\mu \in \text{Spec}(D_y^2)$, the Dirac operator D_y preserves the eigenspace $E_{\mu,y}^p$, indeed, if $s \in E_{\mu,y}^p$, $D_y^2(D_y s) = D_y(\mu s) = \mu(D_y s)$. Since D_y is auto-adjoint, under a basis changing, we can consider the space $\text{Vect}(s)$ generated by s is stable w.r.t. D_y . Thus, $D_y s = \sqrt{\mu} s$ or $D_y s = -\sqrt{\mu} s$. For $s \in E_{\mu,y}^p$ ($\mu \neq b$ because $y \in U^b$),

$$\frac{1}{2\pi i} \int_{|\lambda|=\sqrt{b}} (\lambda - D_y)^{-1} s d\lambda = \frac{1}{2\pi i} \int_{|\lambda|=\sqrt{b}} (\lambda \pm \sqrt{u})^{-1} s d\lambda = \begin{cases} s, & \text{if } 0 < \mu < b \\ 0, & \text{if } \mu > b \end{cases}$$

Therefore,

$$P_y^{b,p} = \frac{1}{2\pi i} \int_{|\lambda|=\sqrt{b}} (\lambda - D_y)^{-1} d\lambda.$$

For each fixed y , the kernel of P_y

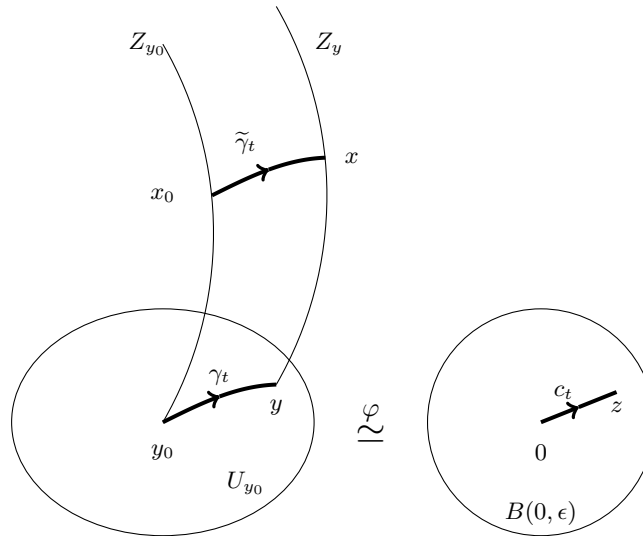
$$P_y(x, x') = \sum_j s_j(x) \otimes s_j(x')^*$$

w.r.t. dv_Z is C^∞ on $x, x' \in Z_y$ where s_j is an orthogonal basis of $K_y^{b,p}$.

Lemma. $P_y(x, x')$ is also C^∞ on $y \in U^b$.

This lemma can be proved by Sobolev Embedding Theorem and Elliptic Estimation. (A DETAILER)

We need a local trivialization of F in help of the parallel transport w.r.t. ∇^F . See the figure below



In a local chart $U_{y_0} \xrightarrow{\varphi} B(0, \epsilon)$ containing y_0 , for each $y \in U_{y_0}$ with $\varphi(y) = z$, take a path $c_t : [0, 1] \rightarrow B(0, \epsilon)$, $t \mapsto tz$. The path c_t corresponds to the path γ_t on U_{y_0} , which can be horizontally lift to $\tilde{\gamma}_t$ in M for any starting

point x_0 . Conversely, since the family of horizontally lifted paths induces a diffeomorphism between the fibre Z_{y_0} and Z_y , for any $x \in Z_y$, we can find a unique lifted path. We get the following identification

$$\begin{array}{ccc} F & & F|_{Z_{y_0}} \\ \downarrow & \cong & \downarrow \\ \pi^{-1}(U_{y_0}) & U_{y_0} \times & Z_{y_0} \end{array}$$

For $y \in U_{y_0}$, $x \in Z_y = \pi^{-1}(y)$, $v \in F|_x$, denote $\tilde{\gamma}_t^x$ the lifted path with target x ,

$$\Phi(y, x, v) = (y, \tilde{\gamma}_0^x, //_{(\tilde{\gamma}^x)^{-1}} v)$$

where $//_{(\tilde{\gamma}^x)^{-1}}$ is the parallel transport along the path $(\tilde{\gamma}^x)^{-1}$ w.r.t. ∇^F . The operator D_y acts on $C^\infty(Z_{y_0}, F|_{Z_{y_0}})$ by

$$D_y s = \Phi \circ D_{y_0} \circ \Phi^{-1} s.$$

Next, we prove that U^b is open. For any $y_0 \in U^b$, take a local chart U_{y_0} as above. Consider the heat operators $e^{-tD_{y_0}^2}$ and $e^{-tD_y^2}$ for $t > 0$. Clearly,

$$\begin{aligned} \text{Spec}\left(e^{-tD_{y_0}^2}\right) &= \left\{e^{-t\mu_j} \mid \mu_j \in \text{Spec}(D_{y_0}^2)\right\} \\ \text{Spec}\left(e^{-tD_y^2}\right) &= \left\{e^{-t\lambda_j} \mid \lambda_j \in \text{Spec}(D_y^2)\right\} \end{aligned}$$

Let $\{s_{y_0}^j\}$ be an orthonormal basis of $C^\infty(Z_{y_0}, F|_{Z_{y_0}})$ formed by eigenvectors corresponding to eigenvalues λ_j . Respectively, $\{s_y^j\}$ be an orthonormal basis of $C^\infty(Z_y, F|_y)$ formed by eigenvectors corresponding to eigenvalues μ_j . As the heat kernel $e^{-tD_y^2}(x, x')$ of $e^{-tD_y^2}$ is C^∞ on x, x' in Z_y , also on $y \in U_{y_0}$. $\forall \epsilon > 0$, we can take U_{y_0} small enough such that $\forall y \in U_{y_0}$,

$$\left\| (e^{-D_y^2} - e^{-D_{y_0}^2}) s \right\|_{L^2} \leq \epsilon \|s\|_{L^2}$$

holds for any $s \in C^\infty(Z_{y_0}, F|_{Z_{y_0}})$.

Take $s = s_y^j$, under the basis $\{s_{y_0}^j\}$, s_y^j can be written as

$$s_y^j = \sum_k a_k s_{y_0}^k.$$

If $y \rightarrow y_0$, $e^{-D_y^2} \rightarrow e^{-D_{y_0}^2}$, thus $\mu_j \rightarrow \lambda_j$ and $s_y^j \rightarrow s_{y_0}^j$. Hence, $a_j \rightarrow 1$ and other coefficients tend to zero. Assume U_{y_0} to be small enough such that $|a_j| > \frac{1}{2}$.

$$\begin{aligned} \epsilon^2 &\geq \left\| (e^{-D_y^2} - e^{-D_{y_0}^2}) s_y^j \right\|_{L^2}^2 = \left\| e^{-\mu_j} s_y^j - \sum_k a_k e^{-\lambda_k} s_{y_0}^k \right\|_{L^2}^2 \\ &= \left\| \sum_k (a_k e^{-\mu_j} - a_k e^{-\lambda_k}) s_{y_0}^k \right\|_{L^2}^2 \\ &\geq \left| a_j e^{-\mu_j} - a_j e^{-\lambda_j} \right| \geq \frac{1}{4} \left| e^{-\mu_j} - e^{-\lambda_j} \right|^2 \end{aligned}$$

So, $\text{Spec}(e^{-D_y^2}) \subset \text{Spec}(e^{-D_{y_0}^2}) +] - 2\epsilon, 2\epsilon[$. Take $\epsilon = \frac{1}{4} \text{dist}(e^{-b}, \text{Spec}(e^{-D_{y_0}^2}))$. Therefore, $e^{-b} \notin \text{Spec}(e^{-D_y^2})$, thus $b \neq \text{Spec}(D_y^2)$, that is $U_{y_0} \subset U^b$. So, U^b is open.

As P_y is C^∞ on y ,

$$\dim K_y^{b,p} \xrightarrow[\text{projection}]{P_y \text{ is}} \text{Tr}[P_y] \xrightarrow[C^\infty \text{ on } x, x']{P_y \text{ is}} \int_{Z_y} \text{Tr}[P_y(x, x)] dv_{Z_y}(x)$$

Hence, $\dim K_y^{b,p}$ is a C^∞ function on y with value in $\mathbb{Z}$. It must be locally constant.

For each $y_0 \in U^b$, take $\{s_{j,y_0}\}$ an orthonormal basis of $K_{y_0}^b$, let us prove that the sections $s_j(y) = P_y \Phi^{-1} s_{j,y_0}$ are still linearly independent for all $y \in U_{y_0}$. Indeed, the linear independance of $\{s_{j,y}\}$ equivalent to the determinant

$$\det \left(\left\langle s_i(y), s_j(y) \right\rangle_{E_y} \right) \neq 0$$

However,

$$\det \left(\left\langle s_i(y), s_j(y) \right\rangle_{E_y} \right) = \det \left(\left\langle P_y \Phi^{-1} s_{i,y_0}, P_y \Phi^{-1} s_{j,y_0} \right\rangle_{E_y} \right)$$

is C^∞ on y . When $y = y_0$, $\det \left(\left\langle P_{y_0} s_{i,y_0}, P_{y_0} s_{j,y_0} \right\rangle_{E_{y_0}} \right) = 1$. Thus, we know that $\{s_{j,y}\}$ are linearly independent. Finally, $\{s_{j,y}\}$ is a basis of K_y^b on U_{y_0} , $\{s_{j,y}\}$ defines a local frame of $K^b|_{U_{y_0}}$. $\square$

For $y \in U^b$, set

$$\lambda_y^b = (\det K_y^{b,0})^{-1} \otimes (\det K_y^{b,1}) \otimes \cdots \otimes (\det K_y^{b,l})^{(-1)^{l+1}}$$

where $\det K_y^{b,j} = \Lambda^{\max} K_y^{b,j}$. Thus, λ^b is a line bundle over U^b . For $0 < b < c$, for $y \in U^b \cap U^c$, define $K_y^{(b,c),p} = \bigoplus_{b < \mu < c} E_{\mu,y}^p$. Since $K^b \oplus K^{(b,c)} = K^c$, $K^{(b,c)}$ is

a vector bundle on $U^b \cap U^c$. Define $\lambda^{(b,c)} = \bigotimes_{p=0}^l (\det K^{(b,c),p})^{(-1)^{p+1}}$ to be a line bundle over $U^b \cap U^c$.

Recall that

$$D^2 = \bar{\partial}^Z (\bar{\partial}^Z)^* + (\bar{\partial}^Z)^* \bar{\partial}^Z.$$

As $\bar{\partial}^Z$ commutes with D^2 , $\bar{\partial}^Z : K^{(b,c),p} \rightarrow K^{(b,c),p+1}$. Denote by $\bar{\partial}^{(b,c)} = \bar{\partial}^Z|_{K^{(b,c)}}$. Thus, we have the following chain complex

$$0 \rightarrow K^{(b,c),0} \xrightarrow{\bar{\partial}^{(b,c)}} K^{(b,c),1} \xrightarrow{\bar{\partial}^{(b,c)}} \cdots \xrightarrow{\bar{\partial}^{(b,c)}} K^{(b,c),l} \rightarrow 0$$

is acyclic.

CONFUSION

IS THE ACYCLIC PROPERTY COME FROM THE D BAR LEMMA?

Here, $\bar{\partial}^{(b,c)} K^{(b,c),j}$, $\bar{\partial}^{(b,c),*} K^{(b,c),j}$ are vector bundles on $U^b \cap U^c$.

Remark. The map $\bar{\partial}^{(b,c)}$ realises each spaces as below:

$$\begin{array}{ccccc} K^{(b,c),j} & = & \bar{\partial}^{(b,c)} K^{(b,c),j-1} & \oplus & \bar{\partial}^{(b,c),*} K^{(b,c),j+1} \\ \downarrow \bar{\partial}^{(b,c)} & & \downarrow \bar{\partial}^{(b,c)} & & \downarrow \bar{\partial}^{(b,c)} \\ K^{(b,c),j+1} & \supset & 0 & \oplus & \bar{\partial}^{(b,c)} K^{(b,c),j} \end{array}$$

The line bundle $\lambda^{(b,c)}$ has a canonical non-zero section

$$T(\bar{\partial}^{(b,c)}) = s_0^{-1} \otimes (\bar{\partial}^{(b,c)} s_0 \otimes s_1) \times (\bar{\partial}^{(b,c)} s_{j-1} \otimes s_j) \otimes \cdots \otimes (\bar{\partial}^{(b,c)} s_{m-1})$$

This section does not depend on the choice of $s_j \in \det(\bar{\partial}^{(b,c),*} K^{(b,c),j+1})$. As the natural direct sum $K^c = K^b \oplus K^{(b,c)}$,

$$\lambda^c = \lambda^b \oplus \lambda^{(b,c)}.$$

Remark. Here, on the intersection domain $U^b \cap U^c$, two line bundles cannot be identified, because the line bundle $\lambda^{(b,c)}$ cannot be ignored or removed. This phenominum is called “Gerbe”.

We identify λ^b and λ^c on $U^b \cap U^c$ by the following C^∞ identification

$$\begin{aligned} \lambda^b &\rightarrow \lambda^c = \lambda^b \otimes \lambda^{(b,c)} \\ s &\mapsto s \otimes T(\bar{\partial}^{(b,c)}) \end{aligned} \quad (*)$$

Definition. We define $\lambda(\xi)$ to be the C^∞ line bundle on B which coincide with λ^b on U^b and $(*)$ is the transition function on $U^b \cap U^c$.

Remark that $\lambda(\xi)$ is C^∞ because $T(\bar{\partial}^{(b,c)})$ is a C^∞ section, because s_j has been chosen C^∞ for the identification.

§3.4 Holomorphic structure on $\lambda(\xi)$

For $0 \leq b \leq c$, denote by $P_y^{(b,c)} : E_y \rightarrow K_y^{(b,c)}$ the orthogonal projection for $y \in U^b \cap U^c$. Recall the Hermitian connection $\tilde{\nabla}$ on E : for $U \in C^\infty(B, TB)$, $s \in C^\infty(B, E) := C^\infty(M, F)$ ($F = \Lambda T^{(0,1)*} Z \otimes \xi$),

$$\tilde{\nabla}_U s = \nabla_{U^H}^F s.$$

Definition. Define a connection $\nabla^b = P^b \widetilde{\nabla}$. Precisely, for $U \in C^\infty(U^b, TU^b)$, $s \in C^\infty(U^b, K^b)$, $\nabla_U^b s = P^b \widetilde{\nabla}_U s$.

Clearly, ∇^b is a Hermitian connection on $(K^b, \langle \cdot, \cdot \rangle_{E|_{K^b}})$ because $\widetilde{\nabla}^E$ is Hermitian. Remark that ∇_U^b preserves $K^{b,j}$ for any $0 \leq j \leq l$. Thus, for any $0 \leq j \leq l$, ∇^b induces a connection on $\det K^{b,j}$ by

$$\nabla_U^b(s_1 \wedge s_2 \cdots \wedge s_{\dim K^{b,j}}) \stackrel{def}{=} \sum_{j=0}^{\dim K^{b,j}} s_1 \wedge \cdots \wedge (\nabla_U^b s_j) \wedge \cdots \wedge s_{\dim K^{b,j}}.$$

Recall the metric on λ^b induced by $\langle \cdot, \cdot \rangle_E$:

$$\text{for } t_1 = t_1^0 \otimes \cdots \otimes t_1^l, t_2 = t_2^0 \otimes \cdots \otimes t_2^l \in \lambda^b = \bigotimes_{j=0}^l \left(\det K^{b,j} \right)^{(-1)^{j+1}},$$

$$\langle t_1, t_2 \rangle_{\lambda^b} = \prod_{j=0}^l \langle t_1^j, t_2^j \rangle_{(\det K^{b,j})^{(-1)^{j+1}}}$$

Proposition. The connection ${}^0\nabla^b$ on λ^b induced by ∇^b preserves the metric $\langle \cdot, \cdot \rangle_{\lambda^b}$.

Proof. For $V \in C^\infty(U^b, TU^b)$, $t_1 = t_1^0 \otimes \cdots \otimes t_1^l, t_2 = t_2^0 \otimes \cdots \otimes t_2^l \in \lambda^b$, (omit the subnote $(\det K^{b,j})^{(-1)^{j+1}}$ of metric)

$$\begin{aligned} V \langle t_1, t_2 \rangle_{\lambda^b} &= V \left(\prod_{j=0}^l \langle t_1^j, t_2^j \rangle \right) \\ &= \sum_{j=0}^l \langle t_1^0, t_2^0 \rangle \cdots \langle t_1^{j-1}, t_2^{j-1} \rangle V(\langle t_1^j, t_2^j \rangle) \langle t_1^{j+1}, t_2^{j+1} \rangle \cdots \langle t_1^l, t_2^l \rangle \\ &= \sum_{j=0}^l \langle t_1^0, t_2^0 \rangle \cdots \langle t_1^{j-1}, t_2^{j-1} \rangle (\langle \widetilde{\nabla}_V t_1^j, t_2^j \rangle) \langle t_1^{j+1}, t_2^{j+1} \rangle \cdots \langle t_1^l, t_2^l \rangle \\ &\quad + \sum_{j=0}^l \langle t_1^0, t_2^0 \rangle \cdots \langle t_1^{j-1}, t_2^{j-1} \rangle (\langle t_1^j, \widetilde{\nabla}_V t_2^j \rangle) \langle t_1^{j+1}, t_2^{j+1} \rangle \cdots \langle t_1^l, t_2^l \rangle \\ &= \langle \widetilde{\nabla}_V t_1, t_2 \rangle_{\lambda^b} + \langle t_1, \widetilde{\nabla}_V t_2 \rangle_{\lambda^b} \\ &= \langle {}^0\nabla_V^b t_1, t_2 \rangle_{\lambda^b} + \langle t_1, {}^0\nabla_V^b t_2 \rangle_{\lambda^b} \end{aligned}$$

□

Theorem. Our previous work satisfies the following fact

- (1) on U^b , there exists an unique holomorphic structure on C^∞ Hermitian line bundle $(\lambda^b, \langle \cdot, \cdot \rangle_{\lambda^b})$ such that ${}^0\nabla^b$ is the Chern connection
- (2) Similary, for $0 < b < c < +\infty$, there exists an unique holomorphic structure on $(\lambda^{(b,c)}, \langle \cdot, \cdot \rangle_{\lambda^{(b,c)}})$ such that ${}^0\nabla^{(b,c)}$ is the Chern connection. And $T(\bar{\partial}^{(b,c)})$ is a non-zero holomorphic section of $\lambda^{(b,c)}$ on $U^b \cap U^c$

(3) holomorphic structures on (λ^b, U^b) patch into a uniquely defined holomorphic structure on $\lambda(\xi)$ on B .

Proof. (1) We need to prove $({}^0\nabla^b)^2$ is of $(1, 1)$ -type. (And then use the Koszul-Malgrange Theorem). By my calculation on Page 27.

$$({}^0\nabla^b)^2 = -\text{Tr}_s[\nabla^{b,2}]$$

CONFUSION

CHECK. Here we have a minus sign, does it mean that my calculation on Page 27 is correct?

Recall the orthogonal projection

$$P^b : E = \Omega^{0,*}(Z, \xi) \rightarrow K^b$$

As the connection ∇^b is on the vector bundle $\begin{array}{c} K^b \\ \downarrow \\ U^b \end{array}$, we can write

$$\nabla^b = P^b \tilde{\nabla} P^b.$$

Denote the orthogonal projection onto the orthogonal complementary space $(K^b)^\perp$ by $P^{(b,+\infty)} = 1 - P^b$, we calculate

$$\begin{aligned} P^b[\tilde{\nabla}, P^b] &= P^b(\tilde{\nabla} P^b - P^b \tilde{\nabla}) = P^b \tilde{\nabla} (1 - P^{(b,\infty)}) - P^b \tilde{\nabla} \\ &= -P^b \tilde{\nabla} P^{(b,\infty)} = \tilde{\nabla} P^b P^{(b,\infty)} - P^b \tilde{\nabla} P^{(b,\infty)} = [\tilde{\nabla}, P^b] P^{(0,\infty)}. \end{aligned}$$

The super-commutator $[\tilde{\nabla}, P^b]$ can be written as $\tilde{\nabla} \cdot P^b$. Thus, $P^b(\tilde{\nabla} \cdot P^b) = (\tilde{\nabla} \cdot P^b) \cdot P^{(0,\infty)}$. This relation tells us that the operator $\tilde{\nabla} \cdot P^b$ interchanges the subspaces K^b and $(K^b)^\perp$. Let us calculate

$$\begin{aligned} \nabla^{b,2} &= (P^b \tilde{\nabla} P^b)^2 = P^b \tilde{\nabla} P^b \tilde{\nabla} P^b \\ &= P^b [P^b \tilde{\nabla} + (\tilde{\nabla} P^b - P^b \tilde{\nabla})] \tilde{\nabla} P^b \\ &= P^b \tilde{\nabla}^2 P^b + P^b [\tilde{\nabla}, P^b] \tilde{\nabla} P^b P^b \\ &= P^b \tilde{\nabla}^2 P^b + P^b [\tilde{\nabla}, P^b] [\tilde{\nabla}, P^b] P^b + \underbrace{P^b [\tilde{\nabla}, P^b] P^b \tilde{\nabla} P^b}_{=0} \\ &= P^b \tilde{\nabla}^2 P^b + P^b [\tilde{\nabla}, P^b] [\tilde{\nabla}, P^b] P^b \\ &= P^b \tilde{\nabla}^2 P^b + P^b (\tilde{\nabla} \cdot P^b) (\tilde{\nabla} \cdot P^b) P^b \end{aligned}$$

Thus,

$$\text{Tr}_s[\nabla^{b,2}] = \underbrace{\text{Tr}_s[P^b \tilde{\nabla}^2 P^b]}_I + \underbrace{\text{Tr}_s[P^b (\tilde{\nabla} \cdot P^b) (\tilde{\nabla} \cdot P^b) P^b]}_{II}.$$

In the previous section, we have proved $\tilde{\nabla}^2 \in \Omega^{1,1}(B, \text{End}(E))$. So, $P^b \tilde{\nabla}^2 P^b \in \Omega^{1,1}(U^b, \text{End}(K^b))$. Then I is of type $(1, 1)$, it rests to prove II is of type $(1, 1)$.

Last week, we proved the equation $[\tilde{\nabla}, \bar{\partial}^Z] = 0$, it implicates

$$[\tilde{\nabla}^{(1,0)}, \bar{\partial}^Z] = [\tilde{\nabla}^{(0,1)}, \bar{\partial}^Z] = 0.$$

Thus,

$$[\tilde{\nabla}^{(0,1)} \cdot P^b, \bar{\partial}^Z] = [[\tilde{\nabla}^{(0,1)}, P^b], \bar{\partial}^Z] = [\tilde{\nabla}^{(0,1)}, \underbrace{[P^b, \bar{\partial}^Z]}_{=0}] - (-1)^{|\tilde{\nabla}^{(0,1)}| |P^b|} [P^b, \underbrace{[\tilde{\nabla}^{(0,1)}, \bar{\partial}^Z]}_{=0}] = 0$$

As the same, $[\tilde{\nabla}^{(1,0)} \cdot P^b, \bar{\partial}^Z] = 0$. Thus,

$$[P^b(\tilde{\nabla}^{(1,0)} \cdot P^b)(\tilde{\nabla}^{(1,0)} \cdot P^b)P^b, \bar{\partial}^Z] = 0.$$

For any $\bar{U}, \bar{V} \in T^{(0,1)}B$,

$$[P^b(\tilde{\nabla}^{(0,1)} \cdot P^b)\tilde{\nabla}^{(0,1)} \cdot P^b P^b, \bar{\partial}^Z](U, V) = [P^b[(\tilde{\nabla}_{\bar{U}}^{(0,1)} \cdot P^b)(\tilde{\nabla}_{\bar{V}}^{(0,1)} \cdot P^b) - (\tilde{\nabla}_{\bar{V}}^{(0,1)} \cdot P^b)(\tilde{\nabla}_{\bar{U}}^{(0,1)} \cdot P^b)]P^b, \bar{\partial}^Z] = 0$$

Lemma. Let $(E, \partial) : 0 \rightarrow E_0 \xrightarrow{\partial} E_1 \xrightarrow{\partial} \dots \xrightarrow{\partial} E_m \rightarrow 0$ be a chain complex of finite dimensional vector space. Let $A = \bigoplus_{j=0}^m \text{End}(E^j)$ satisfying $\partial A = A\partial$, then

$$\sum_{j=0}^m (-1)^j \text{Tr}_{|E_j} [A] = \sum_{j=0}^m (-1)^j \text{Tr}_{|H^j} [A]$$

where $H^j = \frac{\text{Ker} \partial|_{E_j}}{\text{Im} \partial|_{E_{j-1}}}$. Precisely, for $[u] \in H^j$, $A \in \text{End}(H^j)$ is defined by

$A[u] = [Au] \in H^j$. (Proof is omitted)

Apply the lemma, see $P^b[(\tilde{\nabla}_{\bar{U}}^{(0,1)} \cdot P^b)(\tilde{\nabla}_{\bar{V}}^{(0,1)} \cdot P^b) - (\tilde{\nabla}_{\bar{V}}^{(0,1)} \cdot P^b)(\tilde{\nabla}_{\bar{U}}^{(0,1)} \cdot P^b)]P^b$ as A , $\bar{\partial}^Z$ as ∂ ,

$$\begin{aligned} & \text{Tr}_s [P^b(\tilde{\nabla} \cdot P^b)(\tilde{\nabla} \cdot P^b)P^b](U, V) \\ &= \text{Tr}_s [P^b[(\tilde{\nabla}_{\bar{U}}^{(0,1)} \cdot P^b)(\tilde{\nabla}_{\bar{V}}^{(0,1)} \cdot P^b) - (\tilde{\nabla}_{\bar{V}}^{(0,1)} \cdot P^b)(\tilde{\nabla}_{\bar{U}}^{(0,1)} \cdot P^b)]P^b] \\ &= \sum_{j=0}^l (-1)^j \left\{ \text{Tr}_{|H^j} [P^b[(\tilde{\nabla}_{\bar{U}}^{(0,1)} \cdot P^b)(\tilde{\nabla}_{\bar{V}}^{(0,1)} \cdot P^b) - (\tilde{\nabla}_{\bar{V}}^{(0,1)} \cdot P^b)(\tilde{\nabla}_{\bar{U}}^{(0,1)} \cdot P^b)]P^b] \right\} \\ &= \sum_{j=0}^l (-1)^j \left\{ \text{Tr}_{|H^j} [P^b[(\tilde{\nabla}_{\bar{U}}^{(0,1)} \cdot P^b)(\tilde{\nabla}_{\bar{V}}^{(0,1)} \cdot P^b)P^b] - \text{Tr}_{|H^j} [P^b(\tilde{\nabla}_{\bar{V}}^{(0,1)} \cdot P^b)(\tilde{\nabla}_{\bar{U}}^{(0,1)} \cdot P^b)]P^b] \right\} \\ &= 0 \end{aligned}$$

As the same, the $(2, 0)$ -part of $\text{Tr}_s [P^b(\tilde{\nabla} \cdot P^b)(\tilde{\nabla} \cdot P^b)P^b]$ also vanishes. Finally, we obtain $({}^0\nabla^b)^2 \in \Omega^{1,1}(U^b, K^b)$. (2) is similar to (1), omitted.

Next, we prove that $T\left(\bar{\partial}^{(b,c)}\right)$ is a holomorphic section of $\lambda^{(b,c)} \downarrow U^b \cap U^c$. Recall the

acyclic chain complex

$$0 \rightarrow K^{(b,c),0} \xrightarrow{\bar{\partial}^{(b,c)}} K^{(b,c),1} \xrightarrow{\bar{\partial}^{(b,c)}} \dots \xrightarrow{\bar{\partial}^{(b,c)}} K^{(b,c),l} \rightarrow 0$$

And the map $\bar{\partial}^{(b,c)}$ realises each spaces as below:

$$\begin{array}{ccccc} K^{(b,c),j} & = & \bar{\partial}^{(b,c)} K^{(b,c),j-1} & \oplus & \bar{\partial}^{(b,c),*} K^{(b,c),j+1} \\ \downarrow \bar{\partial}^{(b,c)} & & \downarrow \bar{\partial}^{(b,c)} & & \downarrow \bar{\partial}^{(b,c)} \\ K^{(b,c),j+1} & \supset & 0 & \oplus & \bar{\partial}^{(b,c)} K^{(b,c),j} \end{array}$$

Denote $U^{(b,c),j} : K^{(b,c),j} \rightarrow \bar{\partial}^{(b,c),*} K^{(b,c),j+1}$ the orthogonal projection. We have defined a canonical non-zero section

$$T(\bar{\partial}^{(b,c)})$$

on the line bundle $\lambda^{(b,c)}$.

As $\left[\nabla^{(b,c),(0,1)}, \bar{\partial}^{(b,c)} \right] = P^{(b,c)} \left[\tilde{\nabla}^{(0,1)}, \bar{\partial}^{(b,c)} \right] P^{(b,c)} = 0$. Then,

$$\begin{aligned} \nabla^{(b,c),(0,1)} \left((\bar{\partial}^{(b,c)} s_{j-1}) \otimes s_j \right) &= \left(\nabla^{(b,c),(0,1)} (\bar{\partial}^{(b,c)} s_{j-1}) \right) \otimes s_j + (\bar{\partial}^{(b,c)} s_{j-1}) \otimes (\nabla^{(b,c),(0,1)} s_j) \\ &= \left(\bar{\partial}^{(b,c)} \nabla^{(b,c),(0,1)} s_{j-1} \right) \otimes s_j + (\bar{\partial}^{(b,c)} s_{j-1}) \otimes (\nabla^{(b,c),(0,1)} s_j) \end{aligned}$$

The differential form is on base manifold, the operator $\bar{\partial}^{(b,c)}$ is verticle. So,

$$\nabla^{(b,c),(0,1)} \left((\bar{\partial}^{(b,c)} s_{j-1}) \otimes s_j \right) = \left\{ \frac{U^{(b,c),j-1} \nabla^{(b,c),(0,1)} s_{j-1}}{s_{j-1}} + \frac{U^{(b,c),j} \nabla^{(b,c),(0,1)} s_j}{s_j} \right\} (\bar{\partial}^{(b,c)} s_{j-1}) \otimes s_j.$$

Therefore, (s^{-1}, s_l) do not exist

$$\frac{{}^0 \nabla^{(b,c),(0,1)} T(\bar{\partial}^{(b,c)})}{T(\bar{\partial}^{(b,c)})} = \sum_{j=0}^l (-1)^{j-1} \left\{ \frac{U^{(b,c),j-1} \nabla^{(b,c),(0,1)} s_{j-1}}{s_{j-1}} + \frac{U^{(b,c),j} \nabla^{(b,c),(0,1)} s_j}{s_j} \right\} = 0.$$

(3) on $U^b \cap U^c$, the identification

$$\lambda^b \rightarrow \lambda^c, \quad s \mapsto s \otimes T(\bar{\partial}^{(b,c)})$$

is a holomorphic map. Since $K^{c,j} = K^{b,j} \oplus K^{(b,c),j}$,

$$(\det K^{c,j})^{(-1)^{j+1}} = ((\det K^{b,j} \oplus K^{(b,c),j}))^{(-1)^{j+1}} = ((\det K^{b,j} \otimes \det K^{(b,c),j}))^{(-1)^{j+1}} = (\det K^{b,j})^{(-1)^{j+1}} \otimes (\det K^{(b,c),j})^{(-1)^{j+1}}$$

Write $s \otimes T(\bar{\partial}^{(b,c)}) = \bigotimes_{j=0}^l t_j$ with $t_j \in (\det K^{c,j})^{(-1)^{j+1}}$. Write $t_j = s_j \otimes w_j$.

$$\begin{aligned} {}^0 \nabla^c \left(s \otimes T(\bar{\partial}^{(b,c)}) \right) &= \sum_{j=0}^l t_0 \otimes \cdots \otimes (\nabla^c t_j) \otimes \cdots \otimes t_l \\ &= \sum_{j=0}^l t_0 \otimes \cdots \otimes [(\nabla^c s_j) \otimes w_j + s_j \otimes (\nabla^c w_j)] \otimes \cdots \otimes t_l \\ &= \sum_{j=0}^l t_0 \otimes \cdots \otimes [(\nabla^b s_j) \otimes w_j + s_j \otimes (\nabla^{(b,c)} w_j)] \otimes \cdots \otimes t_l \\ &= ({}^0 \nabla^b s) \otimes T(\bar{\partial}^{(b,c)}) + {}^0 \nabla^c s \otimes (\nabla^{(b,c)} T(\bar{\partial}^{(b,c)})) \end{aligned}$$

□

§3.4 Metric and holomorphic connection on $\lambda(\xi)$

Continue the last setting, recall that $|\cdot|^b$ is the L^2 -norm on $\downarrow_{U^b}^{\lambda^b}$, $|\cdot|^{(b,c)}$ is the

L^2 -norm on $\downarrow_{U^b \cap U^c}^{\lambda^{(b,c)}}$. In the previous section, we have proved the coincidence of them. Thus, for $0 < b < c$, on $U^b \cap U^c$, in the identification $\lambda^b \rightarrow \lambda^c$, $s \mapsto s \otimes T(\bar{\partial}^{(b,c)})$,

$$\left| s \otimes T(\bar{\partial}^{(b,c)}) \right|^c = \left| s \right|^b \left| \bar{\partial}^{(b,c)} \right|^{(b,c)}.$$

In the course on 20th Sept 2025, that V is here $D = \bar{\partial}^Z + (\bar{\partial}^Z)^*$, (here $\bar{\partial}^Z$ is that v), that E_j is here $K_j^{(b,c)}$. Here, we do not need to take positive spectrum of D^2 because our setting is $b > 0$, on $K^{(b,c)}$, every eigenvalue is positive. So,

$$\left| T(\bar{\partial}^{(b,c)}) \right|^{(b,c)} = \tau(\bar{\partial}^{(b,c)}) = \prod_{j=0}^l \left(\det D^2|_{K_j^{(b,c)}} \right)^{(-1)^j j/2}$$

In the following, $\theta^{(b,c)}(s)$ (resp. $K^{(b,c)}$) plays the role of $\zeta^{(0)}(s)$ (resp. chain complex E). Omit the variable y , remember the operator D^2 is C^∞ w.r.t. the variable $y \in U^b \cap U^c$. N is the number operator

$$\theta^{(b,c)}(s) = -\text{Tr}_{s|_{K^{(b,c)}}} [N(D^2)^{-s}] = -\text{Tr}_{s|_E} [N(D^2)^{-s} P^{(b,c)}].$$

We have

$$\log \tau^2(\bar{\partial}^{(b,c)}) \log \left\{ \left[\tau(\bar{\partial}^{(b,c)}) \right]^2 \right\} = -\frac{\partial}{\partial s|_{s=0}} \theta^{(b,c)}(s)$$

Remark. We remark here (omit subnote $|_E$ on Tr_s)

$$\theta^{(b,c)}(s) = -\frac{1}{\Gamma(s)} \int_0^\infty u^{s-1} \text{Tr}_s [N e^{-uD^2} P^{(b,c)}] du$$

If we take $c \rightarrow +\infty$, $\dim K^{(b,c)} \rightarrow +\infty$. So, this is the infinite case.

Definition. For $y \in U^b$, $\text{Re}(s) > l = \dim_{\mathbb{C}} Z$. Set

$$\theta_y^b(s) = -\text{Tr}_{s|_E} [N(D^2)^{-s} P^{(b,\infty)}] = -\frac{1}{\Gamma(s)} \int_0^\infty u^{s-1} \text{Tr}_s [N e^{-uD^2} P^{(b,\infty)}] du.$$

Notice that the intergrated part

$$\text{Tr}_s [N e^{-uD^2} P^{(b,\infty)}] = \text{Tr}_s [N e^{-uD^2}] - \text{Tr}_s [N e^{-uD^2} P^b]$$

The term $\text{Tr}_s \left[N e^{-u D^2} P^b \right]$ is exactly $\text{Tr}_{s|_{K^b}} \left[N e^{-u D^2|_{K^b}} P^b \right]$, which is C^∞ w.r.t. u .

Next, let us recall some results of Heat Operator.

Let $F = \Lambda T^{(0,1)*} Z \otimes \xi$. The heat operator is

$$\begin{aligned} e^{-u D^2} : L^2(Z, F) &\rightarrow L^2(Z, F) \\ s &\mapsto e^{-u D^2} s \end{aligned}$$

where $e^{-u D^2} s$ is defined by the unique solution of the following Heat Equation

$$\begin{cases} (\frac{\partial}{\partial u} + D^2)(e^{-u D^2} s) = 0 \\ e^{-u D^2} s \xrightarrow[u \rightarrow 0]{L^2} s \end{cases}$$

Let $e^{-u D^2}(x, x')$ is the Schwartz Kernel of $e^{-u D^2}$ w.r.t $dv_Z(x')$, i.e.

$$e^{-u D^2} s = \int_Z e^{-u D^2}(x, x') s(x') dv_Z(x')$$

We recall the following proposition without proof. See Proof on BGV.

Proposition. *The Heat Kernel $e^{-u D^2}(x, x')$ is C^∞ on $x, x' \in Z_y$ and also on $y \in B$ for $u > 0$. When $u \rightarrow 0$, we have a series expansion for the diagonal $e^{-u D^2}(x, x)$ ($l = \dim_{\mathbb{C}} Z$)*

$$e^{-u D^2}(x, x) \underset{u \rightarrow 0}{=} \sum_{j=-l}^k a_j(x) u^j + o(u^{k+1})$$

uniformly on $x \in Z_y$.

CONFUSION

I DO NOT KNOW EXACTLY WHAT SCHWARTZ KERNEL IS

Remark. *In the Heat Kernel, y belongs to B , not necessarily $y \in U^b$. Because the Heat Kernel can be defined on the total parameter manifold B . When we encounter $\text{Tr}_s \left[N e^{-u D^2} P^b \right]$, the projector P^b forces us to restrict $y \in U^b$. On the other hand, u must be strictly positive, the Heat Equation can only resolve the future event state, cannot go back to the past.*

Proposition. *If $K : L^2(Z, F) \rightarrow L^2(Z, F)$ has a C^∞ kernel w.r.t. dv_Z , then K is a trace class (K can be written to $K = AB$ where A, B are both Hilbert-Schmidt operator) and*

$$\text{Tr}[K] = \int_Z \text{Tr}[K(x, x)] dv_Z(x)$$

From functional analysis, with $\{s_j\}$ an orthonormal basis of $L^2(Z, F)$, we have

$$\mathrm{Tr}[K] = \sum_j \langle K s_j, s_j \rangle_{L^2}.$$

Remark. C^∞ kernel means that $K(x, x')$ not only in sense of distribution, but truly can be expressed by a genuine C^∞ -function. In fact, in our case, the continuity of heat kernel is sufficient to solve the problem, but we still prove here the smoothness of heat kernel.

The heat operator e^{-uD^2} has a C^∞ kernel, although the number operator N take discrete values over different subspaces $\Lambda^j T^{(0,1)} Z \otimes$, but D^2 is stable over each subspace, so the operator $N e^{-uD^2}$ also has C^∞ kernel. By the proposition above, it is trace class and

$$\begin{aligned} \mathrm{Tr}_s [N e^{-uD^2}] &= \int_Z \mathrm{Tr}_s [N e^{-uD^2}(x, x)] dv_Z(x) \\ &\stackrel{\substack{u \rightarrow 0 \\ \text{compact}}}{=} \int_Z \mathrm{Tr}_s [N \sum_{j=-l}^k a_j(x) u^j + o(u^{k+1})] dv_Z(x) \\ &\stackrel{\substack{Z \text{ compact} \\ u \rightarrow 0}}{=} \sum_{j=-l}^k (\mathrm{Tr}_s [N a_j(x)] dv_Z(x)) u^j + o(u^{k+1}). \end{aligned}$$

Thus, if $s > l = \dim_{\mathbb{C}} Z$, the integral

$$\int_0^1 u^{s-1} \mathrm{Tr}_s [N e^{-uD^2}] du$$

is well-defined. Hence,

$$\int_0^1 u^{s-1} \mathrm{Tr}_s [N e^{-uD^2} P^{(b,\infty)}] du = \int_0^1 u^{s-1} \mathrm{Tr}_s [N e^{-uD^2}] - \int_0^1 u^{s-1} \mathrm{Tr}_s [N e^{-uD^2} P^b]$$

is well-defined, the second term is always well-defined for the finiteness of dimension.

For $u \geq 1$,

$$\begin{aligned} &\mathrm{Tr}_{E^j} [e^{-uD^2} P^{(b,\infty)}] \stackrel{\substack{\text{Fun Ana} \\ \text{Definition}}}{=} \sum_k \langle e^{-uD^2} P^{(b,\infty)} s_k, s_k \rangle \\ &= \sum_{\substack{\lambda_j \in \mathrm{Spec}(D^2|_{E^j}) \\ \lambda_j > b}} e^{-u\lambda_j} = \sum_{\substack{\lambda_j \in \mathrm{Spec}(D^2|_{E^j}) \\ \lambda_j > b}} e^{-(u-1)\lambda_j} e^{-\lambda_j} \\ &\leq e^{-(u-1)b} \sum_{\substack{\lambda_j \in \mathrm{Spec}(D^2|_{E^j}) \\ \lambda_j > b}} e^{-\lambda_j} \\ &\leq e^{-(u-1)b} \mathrm{Tr}_{|_{E^j}} (e^{-D^2}) \end{aligned}$$

As e^{-D^2} is trace class, $\mathrm{Tr}_{|_{E^j}} (e^{-D^2})$ is well-defined, then $e^{-(u-1)b} \mathrm{Tr}_{|_{E^j}} (e^{-D^2})$ decreases exponentially to 0 when $u \rightarrow +\infty$. So,

$$\int_1^\infty u^{s-1} \mathrm{Tr}_s [N e^{-uD^2}] du$$

is well-defined.

Definition (Mellin transformation).

Let $f \in C^\infty(]0, +\infty[)$, define the Mellin transformation

$$M[f](s) = \frac{1}{\Gamma(s)} \int_0^\infty f(t)t^{s-1}dt.$$

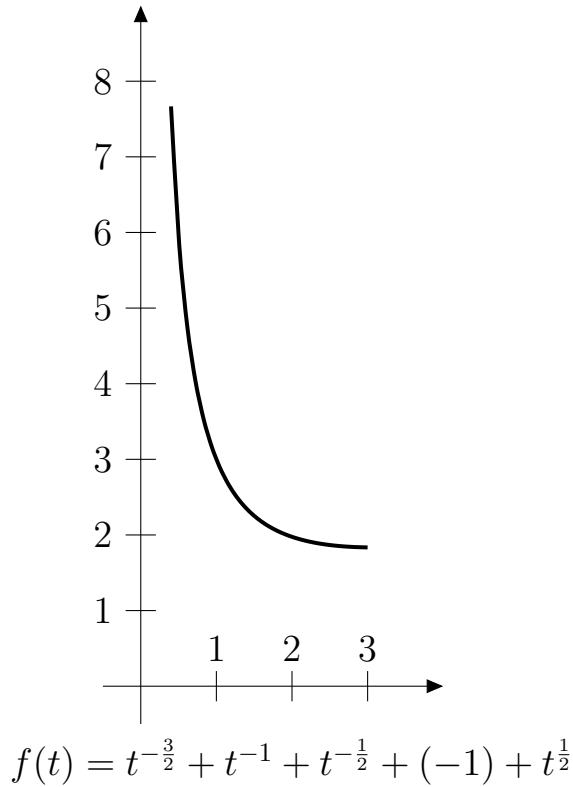
Lemma. Let $f \in C^\infty(]0, +\infty[)$. There exist $C > 0$, $\lambda > 0$ such that

$$(1) \forall k \in \mathbb{N}, f(t) \underset{t \rightarrow 0}{\sim} \sum_{j=-n}^k f_j t^{\frac{j}{2}} + O(t^{\frac{k+1}{2}})$$

$$(2) \forall t > 1, |f(t)| \leq Ce^{-\lambda t}.$$

Then, after Mellin transformation, $M[f]$ is a meromorphic function with poles contains in $\frac{n}{2} - \frac{\mathbb{N}}{2}$. Furthermore, $M[f](s) \underset{s \rightarrow 0}{\sim} f_0 + O(s)$. In particular, $M[f]$ is holomorphic near the origin $s = 0$.

The behavior of $f(t)$ near 0 is figured below.



Proof. For any fixed $k \in \mathbb{N}$,

$$\Gamma(s)M[f](s) = \underbrace{\int_0^1 f(t)t^{s-1}dt}_I + \underbrace{\int_1^\infty f(t)t^{s-1}dt}_{II}.$$

The part II by condtion (2) is well-defined for every $s \in \mathbb{C}$, it is a holomorphic

function. Calculate

$$\begin{aligned}
I &= \underbrace{\int_0^1 [f(t) - \sum_{j=-n}^k f_j t^{\frac{j}{2}}] t^{s-1} dt}_{O(t^{\frac{k+1}{2}+s-1})} + \sum_{j=-n}^k f_j \int_0^1 t^{\frac{1}{2}j+s-1} dt \\
&\quad \underbrace{\hspace{10em}}_{\text{hol for } \operatorname{Re}(s) > -\frac{k+1}{2}} \\
&= \int_0^1 [f(t) - \sum_{j=-n}^k f_j t^{\frac{j}{2}}] t^{s-1} dt + \sum_{j=-n}^k \frac{f_j}{s + \frac{j}{2}}.
\end{aligned}$$

As k is arbitrary and $\frac{1}{\Gamma(s)}$ is an entire function, $M[f]$ is a meromorphic function with poles contains in $\frac{n}{2} - \frac{\mathbb{N}}{2}$.

Furthermore, near the origin, $\frac{1}{\Gamma(s)}$ has an expansion $\frac{1}{\Gamma(s)} \xrightarrow{s \rightarrow 0} s + O(s^2)$.

So the product $\frac{1}{\Gamma(s)} \int_0^\infty f(t) t^{s-1} dt$ will remove the possible pole $s = 0$. This actually makes $M[f]$ be a holomorphic function near the origin. Easy to see that $M[f](s) \xrightarrow{s \rightarrow 0} f_0 + O(s)$. $\square$

Let us calculate $M[f]'(0) = \frac{\partial}{\partial s}|_{s=0} M[f](s)$. Start with derive

$$s\Gamma(s)M[f](s) = s \left[\int_0^1 f(t) t^{s-1} dt + \int_1^\infty f(t) t^{s-1} dt \right].$$

Then,

$$(s\Gamma(s))'M[f](s) + s\Gamma(s)M[f]'(s) = \int_0^1 f(t) t^{s-1} dt + \int_1^\infty f(t) t^{s-1} dt + s \left[\int_0^1 f(t) t^{s-1} dt + \int_1^\infty f(t) t^{s-1} \ln t dt \right]$$

Take $k = -1$ and $s \rightarrow 0$,

$$\begin{aligned}
&\Gamma'(1)f_0 + \Gamma(1)M[f]'(0) \\
&= \int_1^\infty f(t) t^{-1} dt + \lim_{s \rightarrow 0} \left[\int_0^1 \int_0^1 [f(t) - \sum_{j=-n}^{-1} f_j t^{\frac{j}{2}}] t^{s-1} dt + \sum_{j=-n}^{-1} \frac{f_j}{s + \frac{j}{2}} \right] \\
&= \int_1^\infty f(t) t^{-1} dt + \int_0^1 \int_0^1 [f(t) - \sum_{j=-n}^{-1} f_j t^{\frac{j}{2}}] t^{-1} dt + \sum_{j=-n}^{-1} \frac{2f_j}{j}
\end{aligned}$$

As $\Gamma(1) = 1$, $\Gamma'(1) = -\gamma$ (Euler constant). Then,

$$M[f]'(0) = \gamma f_0 + \int_1^\infty f(t) t^{-1} dt + \int_0^1 [f(t) - \sum_{j=-n}^{-1} f_j t^{\frac{j}{2}}] t^{-1} dt + \sum_{j=-n}^{-1} \frac{2f_j}{j}.$$

Back to our problem, the function $u \mapsto \text{Tr}_s[Ne^{-uD^2}P^{(b,\infty)}]$ verifies all conditions in the lemma above, thus

$$\theta_y^b(s) = -\frac{1}{\Gamma(s)} \int_0^\infty u^{s-1} \text{Tr}_s \left[Ne^{-uD^2} P^{(b,\infty)} \right] du$$

can be extended to a meromorphic function on $\mathbb{C}$ and to be holomorphic near the origin $s = 0$.

Definition. For each $y \in U_b$, define

$$\tau_y(\bar{\partial}^{(b,\infty)}) = \exp \left(-\frac{1}{2}(\theta_y^b)'(0) \right).$$

Remark. In the previous text, formally we have

$$\tau(\bar{\partial}^{(b,\infty)}) = \prod_{j=0}^l \left(\det D^2|_{K_j^{(b,\infty)}} \right)^{(-1)^j/2}.$$

But the formal definition does not make a precise sense. The definition above is a strict definition.

Let

$$\|\cdot\|^b = |\cdot|^b \tau(\bar{\partial}^{(b,\infty)})$$

be the metric on the line bundle $\downarrow_{U^b}$. Let

$${}^1\nabla^b = {}^0\nabla^b + \partial^B \log \tau^2(\bar{\partial}^{(b,\infty)})$$

be the connection on the line bundle $\downarrow_{U^b}$.

Recall that if (L, h^L) is a holomorphic Hermitian line bundle, s is a holomorphic local frame of L , then the curvature of the Chern connection ∇^L is

$$R^L = \bar{\partial}\partial \log(|s|_{h^L}^2)$$

We give a proof. For $\bar{U} \in C^\infty(B, T^{(0,1)}B)$, $\nabla_{\bar{U}}^L s = 0$. For $U \in C^\infty(B, T^{(1,0)}B)$,

$$\langle \nabla_U^L s, s \rangle + \langle s, \nabla_{\bar{U}}^L s \rangle = U \langle s, s \rangle = U(|s|_{h^L}^2).$$

Write $\nabla_U^L s = a(u)s$. Then

$$a(u) = (|s|_{h^L}^2)^{-1} U(|s|_{h^L}^2) = U(\log |s|_{h^L}^2).$$

So, $\nabla^L s = \partial \log(|s|_{h^L}^2) \cdot s$. (We must keep s because s must be holomorphic)

$$\begin{aligned} (\nabla^L)^2 s &= \nabla^L \left(\partial \log(|s|_{h^L}^2) \right) \\ &= \left[d(\partial \log(|s|_{h^L}^2)) \right] \cdot s - (\partial \log(|s|_{h^L}^2)) \wedge (\partial \log(|s|_{h^L}^2)) \cdot s \\ &= \left[\bar{\partial} \partial \log(|s|_{h^L}^2) \right] \cdot s \end{aligned}$$

Therefore, on the trivialisation by s ,

$$\nabla^{(L, h^L)} = d + \partial \log(|s|_{h^L}^2).$$

If h^L is replaced by qh^L for a positive function $q \in C^\infty(B,]0, +\infty[)$, then

$$\nabla^{(L, qh^L)} = \nabla^{L, h^L} + \partial \log(q).$$

Theorem. The metric (or norm) $\|\cdot\|^b$ on the line bundle λ^b patch into a smooth metric $\|\cdot\|$ on the line bundle $\lambda(\xi)$ and ${}^1\nabla^b$ patch into a connection ${}^1\nabla$ on $\lambda(\xi)$. The connection ${}^1\nabla$ is exactly the Chern connection on $(\lambda(\xi), \|\cdot\|)$.

Proof.

$$\left\| s \otimes T(\bar{\partial}^{(b,c)}) \right\|^c = |s|^b \underbrace{\tau(\bar{\partial}^{(b,c)}) \tau(\bar{\partial}^{(c,\infty)})}_{\tau(\bar{\partial}^{(b,\infty)})} = \|s\|^b.$$

This means the metric $\|\cdot\|^b$ patch into a C^∞ -metric $\|s\|$ on $\lambda(\xi)$. As ${}^0\nabla^b$ is the Chern connection on $(\lambda^b, |\cdot|^b)$, ${}^1\nabla^b$ is the Chern connection on $(\lambda^b, \|\cdot\|^b)$.

Since $\begin{matrix} (\lambda(\xi), \|\cdot\|) \\ \downarrow \\ B \end{matrix}$ is exactly $\begin{matrix} (\lambda^b, \|\cdot\|^b) \\ \downarrow \\ U^b \end{matrix}$ if restricting on U^b . Thus, ${}^1\nabla^b$ patch together into a connection ${}^1\nabla$ on $\lambda(\xi)$ which is the Chern connection on $(\lambda(\xi), \|\cdot\|)$. $\square$

This metric $\|\cdot\|$ is honored to be called “Quillen Metric”.

Next, we want to calculate explicitly this amount on a local chart.

Let $A_u = \tilde{\nabla} + \sqrt{u}D$ be an operator on $E = \Omega^{0,*}(Z, \xi)$. The zero-degree part of $e^{-A_u^2}$ is e^{-uD^2} . Recall some knowledge.

Let (Z, g^{TZ}) be a compact Riemannian manifold, (F, h^F) be a Hermitian vector bundle on Z with Hermitian connection ∇^F . Set

$$H = \Delta^F + Q$$

where $\Delta^F = -\sum_j \left[(\nabla_{e_j}^F)^2 - \nabla_{\nabla_{e_j}^{TX} e_j}^F \right]$ is the Laplacian associated to ∇^F , $Q \in C^\infty(Z, \text{End}(F))$ is Hermitian. Let $\mathcal{A} = \sum_{j=0}^n \mathcal{A}^j$ (consider to be the exterior algebra over $\mathbb{R}$) be a finite dimensional $\mathbb{Z}$ -grading $\mathbb{R}$ -algebra ($\mathcal{A}^i \cdot \mathcal{A}^j \subset \mathcal{A}^{i+j}$). Denote by $\mathcal{A}^{>0} = \sum_{j>0} \mathcal{A}^j$. Set an operator on $L^2(X, F) \otimes \mathcal{A}$,

$$\mathcal{F} = H_0 + K + \mathcal{F}_+$$

where

$$\begin{aligned} H_0 : L^2(Z, F) \otimes \mathcal{A}^0 &\rightarrow L^2(Z, F) \otimes \mathcal{A}^0 \\ K : L^2(Z, F) \otimes \mathcal{A} &\rightarrow L^2(Z, F) \otimes \mathcal{A} \\ \mathcal{F}_+ : L^2(Z, F) \otimes \mathcal{A}^{>0} &\rightarrow L^2(Z, F) \otimes \mathcal{A}^{>0} \end{aligned}$$

CONFUSION

FRANKLY THIS PART I HAVE NOT WELL FOLLOW THE PROF. THE PROF JUST INSTRUCTED TOO QUICKLY.

Assume that the disturbance operator K has a C^∞ kernel.

We will define the kernel $P_t(x, x') \in F_x \otimes F_{x'}^* \otimes \mathcal{A}$, that is to be the solution of

$$\begin{cases} \left(\frac{\partial}{\partial t} + \mathcal{F}_x \right) P_t(x, x') = 0 \\ \int_{x' \in Z} P_t(x, x') s(x') dv_Z(x') \xrightarrow{t \rightarrow 0} s \text{ uniformly on } x \in Z \end{cases}$$

We start from e^{-tH_0} which is obtained by using volterra series. At first, take $\mathcal{A} = \mathbb{R}$. We wish to construct the Heat kernel $e^{-tH}(x, x')$ for $H = H_0 + K$. Temporarily, we leave the nilpotent term $\mathcal{F}_+$.

Set the k -simplex in $\mathbb{R}^{k+1}$

$$\Delta_k = \left\{ (\sigma_0, \sigma_1, \dots, \sigma_k) \mid \sigma_0 + \dots + \sigma_k = 1 \right\}$$

Under the variable changing

$$\sigma_0 = t_1, \sigma_1 = t_2 - t_1, \dots, \sigma_k = 1 - t_k$$

we have

$$\Delta_k = \left\{ (t_1, \dots, t_k) \in \mathbb{R}^k \mid 0 \leq t_1 \leq t_2 \leq \dots \leq t_k \leq 1 \right\}.$$

Set the operator

$$Q_{k,t} = \int_{\Delta_k} e^{-\sigma_0 t H_0} K e^{-\sigma_1 t H_0} K \dots K e^{-\sigma_k t H_0} d\sigma_0 d\sigma_1 \dots d\sigma_k$$

Proposition. Consider the kernel $Q_{k,t}(x, x')$ of the operator $Q_{k,t}$. Then,

$$\sum_{k=0}^N (-t)^k Q_{k,t}(x, x') \xrightarrow{C^l \xrightarrow{N \rightarrow \infty}} q_t(x, x') \in C^\infty(Z \times Z, \pi_1^* F \otimes \pi_2^* F)$$

where π_1, π_2 are canonical projections on first and second component. Furthermore, $q_t(x, x')$ is C^∞ on t and satisfies

$$\begin{cases} (\frac{\partial}{\partial t} + H_x)q_t(x, x') = 0 \\ \int_{x' \in Z} q_t(x, x')s(x')dv_Z(x') \xrightarrow{t \rightarrow 0} s \text{ uniformly on } x \in Z \end{cases}$$

Thus, q_t is the Heat kernel for $H = H_0 + K$ and $(e^{-tH} - e^{-tH_0})(x, x') \xrightarrow{t \rightarrow 0} 0$

* * *